PROJECT CASE STUDY

SAFISH PLUS DISTRIBUTOR ERP

A desktop trading and distribution system connecting fish purchases, stock sales, commission consignments, vendor settlements, and customer accounts in one English–Urdu workspace.

SYSTEM	SCOPE	STATUS
Fish Wholesale & Distribution Management	**Trading · Stock · Accounts Commission Settlements**	**Implemented Desktop Application**

01 The Problem

Fish wholesalers and commission agents need to track more than a sale price. Each trading day connects variable fish weights, purchase rates, customer collections, transport and ice charges, vendor advances, and settlement deductions.

The same business can sell its own stock and sell consignments on behalf of vendors. Those two flows have different inventory and settlement effects; treating them as one makes the day’s stock position and outstanding accounts harder to follow.

> The application brings purchases, sales, inventory movement, customer accounts, vendor bills, and operational reports into a single locally stored desktop workflow.

02 What Was Built

A bilingual desktop management system for fish traders, distributors, and commission agents, with persistent business records rather than browser-only demonstrations.

TRADING & INVENTORY

Record fish purchases and weight-based sales. Owned-stock transactions increase or reduce available inventory.

CUSTOMER & VENDOR ACCOUNTS

Maintain customer and vendor records, opening balances, transaction balances, and displayed vendor advances.

COMMISSION SETTLEMENTS

Build vendor bills from eligible unbilled consignment sales, with commission, charges, concessions, and cash paid.

REPORTING & DOCUMENT OUTPUT

Explore thirteen query-backed report views, generate PDF/Excel output, and review trading and account history.

03 How It Works

Users enter and review transactions in the desktop workspace. Domain requests cross the preload bridge to main-process handlers, which validate inputs and read or update a local SQLite database.

React workspace → Preload bridge → IPC handlers → SQLite

Posted purchases + stock sales → Inventory movements and account balances

Unbilled consignments → Commission settlement → Linked vendor bills

Purchases & owned stock

A purchase records its vendor, date, fish, weight, rate, concession, and payment. Forms convert between kg and maund rates (1 maund = 40 kg). Saving posts the purchase, increases item stock, and adds the unpaid amount to the vendor’s balance.

Sales & customer accounts

Sale amounts are calculated from weight and rate. Fare and ice charges contribute to the net amount; cash and receipts reduce the outstanding amount. Only stock-sourced sale lines reduce owned inventory.

Consignment & vendor settlement

A bill preview selects a vendor’s posted, non-stock sale lines that have not already been billed. The bill deducts commission, charges, and concession, records cash paid, and links the included lines to the settlement.

Review, reports & output

Search screens retrieve stored records for review and correction. The dashboard shows counts, stock, and vendor advances; reports query the same transaction history for account, stock, sales, and purchase views.

04 Key Features

- **Weight-based sales and purchases** — fish lines retain weight, rate, amount, and relevant charges, with document numbers and posted transaction records.

- **Owned stock and vendor consignments** — stock-sourced sales reduce inventory; non-stock sales remain available for a separate commission-billing workflow.

- **Customer and vendor management** — searchable master records connect identity, location, contact details, opening balances, and trading history.

- **Vendor bill generation** — inclusive date-range previews select eligible sales and calculate commission, expenses, concession, payment, and remaining settlement balance.

- **Thirteen operational reports** — customer recovery, ledger, customer register, concession, net summary, daily sales/details, item sales/purchases, vendor sales/stock bill, stock, and stock-sale history.

- **PDF and Excel output** — generate and open PDF documents through Electron, or save structured XLSX data through ExcelJS. Reports provide their own printable document layouts.

- **English–Urdu desktop experience** — localized labels and right-to-left presentation work alongside desktop navigation, search controls, keyboard shortcuts, and light/dark themes.

- **Search and transaction correction** — date and entity filters retrieve saved sales and purchases. Edit and delete paths reverse and reapply their linked stock and account effects.

05 Technical Approach

The implementation separates the renderer interface, the preload API, main-process business requests, and local data persistence. Electron supplies the desktop runtime; Vite builds the React renderer.

LAYER	STACK	PURPOSE
Desktop runtime	Electron 40 · Node.js	Window lifecycle, preload bridge, IPC handlers, native dialogs, and desktop output.
Frontend & tooling	React 19 · JavaScript · Vite 7	Component-based screens, local UI state, renderer development, and production bundling.
State & styling	Zustand · Mantine 8 · Tailwind CSS 4	Application settings, domain screens, forms, shared components, and theme-aware styling.
Data persistence	SQLite · better-sqlite3	Local records, prepared queries, transactional creates, foreign keys, and WAL-mode operation.
Localization	i18next · react-i18next	English/Urdu catalogs, language switching, right-to-left presentation, and key-coverage checking.
Output & packaging	Electron printToPDF · ExcelJS · electron-builder	PDF and spreadsheet output, print handling, and desktop packaging configuration.

Design decisions that matter

A narrow renderer-to-main boundary. The UI calls domain-specific methods exposed through a context-isolated preload bridge instead of issuing renderer-side SQL.

Linked records and transactional writes. Sales, purchase lines, stock updates, and balances are maintained together in create transactions; vendor bills retain links to their included consignment lines.

Explicit desktop scope. The repository is a local trading-management application, not a cloud service or complete double-entry accounting suite. Configuration screens do not by themselves establish live external integrations.

06 Value Delivered

From separate trading records to a connected, inspectable desktop workflow:

Connected	Traceable	Bilingual
TRADING WORKFLOWS	LINKED RECORDS	LOCAL OPERATIONS

- ✓ **One operational workspace.** — Purchases, sales, contacts, stock, settlement bills, and reports share an application and a local database rather than isolated screen-only data.
- ✓ **Clearer ownership and settlement flows.** — Owned inventory is distinct from vendor consignment sales, while eligible sale lines remain linked to the bills that settle them.
- ✓ **Inspectable balances and trading activity.** — Customer recovery, account ledgers, registers, stock reports, and day-end summaries make the underlying transaction history available for review.
- ✓ **A practical local-language interface.** — English and Urdu labels, right-to-left support, native desktop output, and locally stored records suit a locally operated fish-trading workflow.

SAFISH PLUS DISTRIBUTOR ERP — a local-first desktop trading system connecting inventory movements, customer accounts, commission settlements, and operational reporting for fish wholesalers.